

University of Kentucky

College of Education — Department of Educational Technology

TEK 200 — Introduction to Educational Technology Spring 2026 Syllabus

Instructor: Dr. Sarah Chen

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Office Hours: Tuesdays & Thursdays 2:00 – 3:30 PM, or by appointment

Meeting Times: Mon/Wed/Fri 10:00 – 10:50 AM

Location: Taylor Education Building 206

Semester: January 13 – April 25, 2026 (15 weeks)

Prerequisites

TEK 100 — Foundations of Teaching and Learning (required). Students must have completed TEK 100 prior to enrollment. This course builds on the pedagogical frameworks, assessment basics, and classroom management principles covered in TEK 100.

Course Description

This course provides a comprehensive introduction to educational technology and its role in modern teaching and learning. Students will explore how artificial intelligence, learning management systems, multimedia tools, and data-driven approaches are transforming K-20 education. Through hands-on projects and critical analysis, students will develop practical skills in selecting, evaluating, and integrating technology into instructional practice while maintaining ethical and equitable standards.

Learning Objectives

By the end of this course, students will be able to:

1. Evaluate the pedagogical effectiveness of emerging educational technologies, including AI-powered tools, using evidence-based frameworks.
2. Design a technology-enhanced lesson plan that aligns learning objectives, assessments, and instructional activities using backward design principles.
3. Analyze the ethical implications of AI and data analytics in educational settings, including issues of bias, privacy, and digital equity.
4. Create multimedia learning materials that apply principles of cognitive load theory and universal design for learning (UDL).
5. Synthesize current research on educational technology to develop a position paper advocating for responsible technology integration in a specific educational context.
6. Apply formative and summative assessment strategies that leverage technology to provide timely, actionable feedback to diverse learners.

Required Texts & Materials

- Bates, A.W. (2025). Teaching in a Digital Age (4th ed.). Open textbook — free online.
- Selwyn, N. (2024). Education and Technology: Key Issues and Debates (4th ed.). Bloomsbury.
- Additional readings posted weekly on Canvas.

Grading Breakdown

Participation & Discussion Posts: 15%
Weekly Assignments (5 total): 40%
Midterm Project — Technology Audit: 20%
Final Project — Integration Proposal: 25%

Grading Scale: A (90-100%), B (80-89%), C (70-79%), D (60-69%), E (below 60%)

Course Policies

Attendance Policy

Regular attendance is expected. More than three unexcused absences will result in a 5% deduction from your final grade per additional absence. If you must miss class, notify the instructor in advance.

Late Policy

Assignments submitted late will lose 10% per calendar day, up to a maximum of 3 days. After 3 days, late work will not be accepted unless a prior extension has been approved. Technology failures are not accepted as excuses — submit early.

Academic Integrity

All work must be your own. Use of AI writing tools must be disclosed and is permitted only for brainstorming and outlining, not for final submissions. Violations will be reported to the Office of Academic Ombud Services per University Senate Rules 6.3.1.

Course Schedule

Week 1: Introduction: What Is Educational Technology?

Dates: January 13 – January 17

Readings:

- Bates Ch. 1: Fundamental Change in Education
- Selwyn Ch. 1: Technology and Education — An Introduction

Discussion: Share your earliest memory of technology in a classroom. What worked? What didn't?

Week 2: Learning Theory Meets Technology

Dates: January 20 – January 24

Readings:

- Bates Ch. 2: The Nature of Knowledge and Epistemology
- Clark, R. (1994) "Media Will Never Influence Learning" (Canvas)

Assignment 1: Analyze a learning app through the lens of behaviorism, cognitivism, and constructivism. Due: January 24, 2026.

Week 3: Backward Design and Alignment

Dates: January 27 – January 31

Readings:

- Wiggins & McTighe (2005) Understanding by Design, Ch. 1 (Canvas)
- Bates Ch. 4: Methods of Teaching with an Online Focus

Discussion: Critique a real syllabus for alignment gaps between objectives, activities, and assessments.

Week 4: Learning Management Systems and Digital Ecosystems

Dates: February 3 – February 7

Readings:

- Bates Ch. 6: Understanding Technology in Education
- Selwyn Ch. 4: Education and the Internet

Assignment 2: Build and populate a sample Canvas module for a unit of your choice. Due: February 7, 2026.

Week 5: Multimedia Learning and Cognitive Load Theory

Dates: February 10 – February 14

Readings:

- Mayer, R. (2021) Multimedia Learning, Ch. 3 (Canvas)
- Bates Ch. 7: Pedagogical Roles for Audio and Video

Discussion: Evaluate three educational videos for Mayer's multimedia principles. Post findings to discussion board.

Week 6: Universal Design for Learning (UDL)

Dates: February 17 – February 21

Readings:

- CAST (2024) UDL Guidelines 3.0 (online)
- Selwyn Ch. 6: Digital Inequalities and Education

Assignment 3: Redesign an inaccessible learning resource using UDL principles. Due: February 21, 2026.

Week 7: Midterm Project Workshop — Technology Audit

Dates: February 24 – February 28

Readings:

- Review all readings from Weeks 1-6

In-class workshop: Peer review drafts of your Technology Audit. Midterm Project due: February 28, 2026.

Week 8: Introduction to AI in Education

Dates: March 3 – March 7

Readings:

- Holmes, W. et al. (2023) AI and Education, Ch. 2-3 (Canvas)
- Bates Ch. 12: Planning for the Future

Discussion: Hands-on exploration of three AI-powered educational tools. Compare affordances and limitations.

Week 9: Spring Break — No Classes

Dates: March 10 – March 14

No assignments due.

Week 10: AI Ethics, Bias, and Equity in EdTech

Dates: March 17 – March 21

Readings:

- Selwyn Ch. 8: Critical Perspectives on EdTech
- O'Neil, C. (2016) Weapons of Math Destruction, Ch. 3 (Canvas)

Assignment 4: Write an ethical impact assessment for an AI tool used in K-12 education. Due: March 21, 2026.

Week 11: Data-Driven Decision Making and Learning Analytics

Dates: March 24 – March 28

Readings:

- Siemens, G. (2013) "Learning Analytics" (Canvas)
- Bates Ch. 10: Trends in Open Education

Discussion: Examine a sample learning analytics dashboard. What decisions can it support? What risks does it create?

Week 12: Assessment Innovation: Portfolios, Badges, and Rubrics

Dates: March 31 – April 4

Readings:

- Mueller, J. (2024) Authentic Assessment Toolbox (online)
- Selwyn Ch. 9: The Future of Education and Technology

Assignment 5: Design a rubric and portfolio framework for a course of your choosing. Due: April 4, 2026.

Week 13: Emerging Technologies: VR, AR, and Immersive Learning

Dates: April 7 – April 11

Readings:

- Radianti, J. et al. (2020) "Systematic Review of VR in Education" (Canvas)
- Bates Ch. 8: Emerging Technologies

Discussion: After trying a VR classroom demo, evaluate its potential for your discipline.

Week 14: Responsible Innovation and the Future of EdTech

Dates: April 14 – April 18

Readings:

- Holmes, W. et al. (2023) AI and Education, Ch. 7: Policy (Canvas)
- Selwyn Ch. 10: Toward More Critical EdTech

Final Project workshop: Present a 5-minute pitch of your Integration Proposal for peer feedback.

Week 15: Final Presentations and Course Wrap-Up

Dates: April 21 – April 25

Final Project — Technology Integration Proposal due: April 25, 2026. In-class presentations during final exam slot.

Assignment Summary with Due Dates

- Assignment 1: Learning Theory App Analysis — Due: January 24, 2026
- Assignment 2: Canvas Module Build — Due: February 7, 2026
- Assignment 3: UDL Resource Redesign — Due: February 21, 2026
- Midterm Project: Technology Audit — Due: February 28, 2026
- Assignment 4: AI Ethical Impact Assessment — Due: March 21, 2026
- Assignment 5: Rubric & Portfolio Framework — Due: April 4, 2026
- Final Project: Technology Integration Proposal — Due: April 25, 2026